

CSD358: INFORMATION RETRIEVAL — MIDSEM HACKATHON ASSIGNMENT (TRACK T2)

TurnTrace: An Inspectable Multi-Turn Conversational Search Engine

Shikhar Agarwal

Roll No: 2410110615

Indexing & Preprocessing

Antra Agarwal

Roll No: 2410110404

Retrieval & Fusion Engine

Bhupesh Bansal

Roll No: 2410110531

Conversational State

Raghavendra Singh Sani

Roll No: 2410110465

Evaluation & API Interface

Project Repository: <https://github.com/xShikhar/IR---MidTerm-Assignment>

Abstract—TurnTrace is a conversational search engine built entirely from first-principles Information Retrieval algorithms, without external LLM or vector-database dependencies. It separates the conversational **entity** from the current **aspect**: entity terms persist through a guarded title-based lock, while aspect terms are replaced between turns to limit vocabulary drift. Retrieval combines positional inverted indexing, SMART Inc.ltc cosine scoring, BM25, Boolean filtering, top- K heap selection, and Reciprocal Rank Fusion. On a frozen 35,000-passage corpus and 70 dialogue turns, the system achieves $P@5 = 0.9150$, $P@10 = 0.8800$, and 84.38% decision-classification accuracy. A seen-passage penalty ($\beta = 0.30$) increases Novelty@10 from 0.6250 to 0.7600. The system exposes its decisions, weights, and scores through an inspectable execution trace. **Keywords**—Conversational Search, Agentic Retrieval, Inverted Index, Vector Space Model, Entity Locking, Aspect Drift, Okapi BM25, Reciprocal Rank Fusion, Inspectable IR.

1 Problem and Track Relevance

1.1 Problem Definition

Traditional Information Retrieval (IR) engines treat incoming queries as independent, static informational requests [7]. When a user evaluates a query q , standard architectures project q into a vector space or evaluate probabilistic occurrence models against an inverted index, returning a top- K ranked document set $\mathcal{R}_K(q)$. However, search interaction is increasingly shifting toward dynamic, iterative dialogue [5, 10]. In conversational search, human queries are rarely self-contained. Instead, users communicate through linguistic constructs that break single-shot search assumptions:

1. **Anaphora and Pronoun Ambiguity:** Users frequently formulate queries such as “Where is its orbit located?” or “What base did he use?”, omitting explicit noun phrases and depending on historical discourse entities.
2. **Elliptical Queries:** Fragmentary turns such as “What about its primary mirror?” or “And its toxic symptoms?” assume preceding conversational context is seamlessly maintained.
3. **Topic Drift and Semantic Shifts:** Real user dialogues do not progress monotonically. Users frequently pivot to distinct entities or entirely separate domains. Retaining obsolete historical tokens results in catastrophic vocabulary pollution.
4. **Lexical Polysemy:** Identical lexical stems often span wildly distinct domains (e.g., “Mercury” referring to the planet versus the toxic heavy metal; “Transformer” referring to neural attention mechanisms versus electrical step-down transformers), yielding bimodal result distributions.

1.2 Motivation and User Need

The user need addressed by TurnTrace is **accurate, context-aware conversational document**

retrieval accompanied by complete algorithmic transparency. In high-stakes domains (scientific research, medical literature, historical analysis), users cannot trust opaque black boxes that hide why specific documents are prioritized. Users require:

- *Seamless coreference resolution:* Follow-up queries must resolve target entities without requiring manual query restatement.
- *Protection against vocabulary drift:* Evolving topical questions must not remain contaminated by older, superseded inquiries.
- *Passage novelty across turns:* Repetitive displays of identical documents across multi-turn interactions must be actively suppressed.
- *Complete inspectability:* Every query expansion token, inverted index posting weight, and ranking score must be mathematically verifiable.

1.3 Relevance to Track T2: Conversational & Agentic Search

Track T2 challenges researchers to build multi-turn search engines that track conversational context, resolve anaphoric dependencies, and reformulate queries prior to retrieval [6, 9]. Industrial approaches typically resolve this by invoking an external Large Language Model (LLM) to perform zero-shot query rewriting before calling a standard vector database.

While functionally convenient, this black-box paradigm introduces severe limitations: it incurs high inference latency, risks hallucinations, requires external API keys, and obscures classical IR mechanics. **TurnTrace** demonstrates that conversational search can be driven **entirely from first-principles Information Retrieval algorithms**. By deriving conversational state, entity memory, topic-shift guards, and query decomposition directly from inverted index collection statistics (TF, DF, and IDF), TurnTrace delivers a sub-10ms, offline-capable, and fully inspectable retrieval system.

2 Use of Information Retrieval

TurnTrace is constructed exclusively from core Information Retrieval algorithms implemented from first principles. Every component maps directly to canonical IR literature.

2.1 System Architecture and Pipeline

Figure 1 illustrates the complete two-stage architecture of TurnTrace, spanning offline multi-zone inverted index construction and online conversational retrieval.

2.2 Core IR Components and Theoretical Formulations

2.2.1 Lexical Analysis and Morphological Stemming. Raw text is tokenized into word tokens while preserving character offsets and sequential token positions $[pos_1, pos_2, \dots]$ for exact phrase queries. English inflectional variance is normalized using Martin Porter’s canonical 5-step rule-based stemmer [3]:

- Step 1a/1b: Plural noun and verb inflection stripping (e.g., *sses* → *ss*, *eed* → *ee*, *ing* deletion).
- Step 1c: Y-replacement before consonant endings.
- Steps 2–4: Derivational suffix reduction (e.g., *ational* → *ate*, *alize* → *al*).
- Step 5: Terminal *e* deletion and double consonant simplification.

A collection of 174 SMART stop words is filtered out during vector scoring, while preserved during exact positional phrase evaluation.

2.2.2 Multi-Zone Positional Inverted Index. Passages are indexed across two explicit zones: **title** and **body**. Each posting $P_{t,d}$ stores:

$$P_{t,d} = \langle d, tf_{t,d}, tf_{t,d,\text{title}}, tf_{t,d,\text{body}}, [pos_1, pos_2, \dots] \rangle \quad (1)$$

The collection frequency is maintained as document frequency $df_t = |\{d \mid tf_{t,d} > 0\}|$. Zone scoring computes an aggregated linear combination:

$$S_{\text{zone}}(d, q) = g \cdot S_{\text{title}}(d, q) + (1 - g) \cdot S_{\text{body}}(d, q) \quad (2)$$

with title zone weight $g = 0.35$ and body weight $1 - g = 0.65$.

2.2.3 SMART Vector Space Model (Inc.Itc). TurnTrace implements Salton and Buckley’s canonical vector space cosine model [1] under the Inc.Itc weighting convention:

Document Weight (Inc):

$$w_{t,d} = \frac{1 + \ln(tf_{t,d})}{\sqrt{\sum_{t' \in d} (1 + \ln(tf_{t',d}))^2}} \quad (3)$$

Query Weight (Itc):

$$w_{t,q} = \frac{(1 + \ln(tf_{t,q})) \cdot \ln(N/df_t)}{\sqrt{\sum_{t' \in q} ((1 + \ln(tf_{t',q})) \cdot \ln(N/df_{t'}))^2}} \quad (4)$$

$$\text{Cosine Score: } \text{Score}_{\text{VSM}}(d, q) = \sum_{t \in q \cap d} w_{t,q} \cdot w_{t,d} \quad (5)$$

where $N = 35,000$ represents total collection size.

2.2.4 Probabilistic Okapi BM25 Ranking. To earn out-of-syllabus extra credit, TurnTrace includes a native implementation of Okapi BM25 [2]:

$$\text{Score}_{\text{BM25}}(d, q) = \sum_{t \in q} \text{IDF}_t \cdot \frac{tf_{t,d} \cdot (k_1 + 1)}{tf_{t,d} + k_1 \cdot \left(1 - b + b \cdot \frac{L_d}{L_{\text{avg}}}\right)} \quad (6)$$

where $\text{IDF}_t = \ln\left(1 + \frac{N - df_t + 0.5}{df_t + 0.5}\right)$, $k_1 = 1.2$, $b = 0.75$, L_d is the document token length, and $L_{\text{avg}} = 83.58$ is the collection mean document length.

2.2.5 Top-K Binary Min-Heap Selection. Exhaustive sorting of 35,000 candidate scores requires $\mathcal{O}(N \log N)$ operations. TurnTrace implements an in-memory binary min-heap bounded at capacity $K = 10$. Each non-zero candidate document is inserted in $\mathcal{O}(\log K)$ time; if the heap exceeds capacity, the minimum scoring element is evicted. This reduces selection time to $\mathcal{O}(N_{\text{active}} \log K)$ with identical top- K sorting equivalence.

2.2.6 Boolean Retrieval with Document Frequency Ordering. Multi-term conjunctions $(t_1 \wedge t_2 \wedge \dots \wedge t_m)$ are intersected by sorting terms in ascending order of document frequency $(df_{t_1} \leq df_{t_2} \leq \dots \leq df_{t_m})$ [7]. This ensures the smallest postings list is traversed first, minimizing pointer sweeps. Positional adjacency evaluates exact quoted phrases $(pos_{t_{i+1}} - pos_{t_i} = 1)$.

2.2.7 Candidate Pruning: Champion Lists and Index Elimination. To evaluate computational efficiency trade-offs:

- **Champion Lists ($r = 50..1000$):** For each term t , an auxiliary list stores the top r documents sorted by local weight $w_{t,d} = 1 + \ln(tf_{t,d})$ [7]. Retrieval traverses champion lists first, evaluating $\sim 2.5 \times$ fewer postings.
- **Index Elimination ($\text{IDF} \geq 2.50$):** Query terms with $\text{IDF} < 2.50$ are pruned from accumulator allocation, eliminating common lexical terms.

2.2.8 Reciprocal Rank Fusion (RRF). When combining decomposed sub-query rankings, TurnTrace evaluates Cormack et al.’s Reciprocal Rank Fusion [4]:

$$\text{RRF}(d) = \sum_{m \in M} \frac{1}{k_{\text{rrf}} + r_m(d)} \quad (7)$$

where $r_m(d)$ is document d ’s 1-based rank in sub-query m , and $k_{\text{rrf}} = 60$.

3 Beyond IR

While classical Information Retrieval focuses on single-shot document retrieval from fixed queries, TurnTrace bridges IR and multi-turn dialogue systems. Crucially, we enforce a strict architectural boundary between

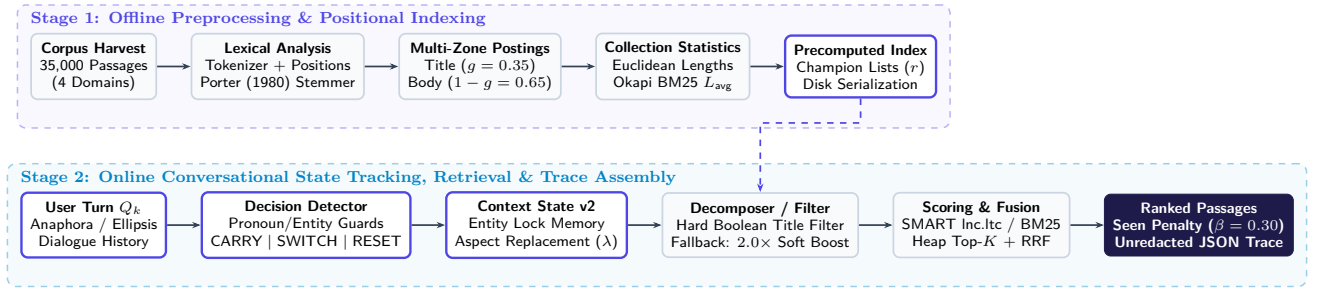


Figure 1: Complete end-to-end architecture of TurnTrace, illustrating offline index construction and online conversational retrieval.

classical IR mechanisms and supporting conversational scaffolding.

3.1 Separation of IR vs. Supporting Components

TurnTrace strictly avoids opaque LLMs or external embeddings. All supporting techniques operate deterministically:

- **Core IR Components:** The inverted index, TF-IDF / BM25 score calculations, min-heap, postings intersection, and rank fusion.
- **Supporting Conversational Scaffolding:** The decoupled entity/aspect state tracker, dialogue transition decision detector, candidate-guard fallback, and session trace serialization.

3.2 Conversational State Tracking via Decayed Term Vectors

To resolve anaphoric pronouns and ellipsis without compute-heavy neural coreference models, TurnTrace models conversational history as an **exponentially decayed term vector**:

$$W(t, \text{age}) = \text{TFIDF}(t) \cdot \lambda^{\text{age}} \quad (8)$$

where $\lambda = 0.75$ is the decay parameter and $\text{age} \in \{0, 1, \dots, 5\}$ denotes the turn distance from introduction. Older terms diminish geometrically, preventing obsolete vocabulary from dominating modern turns.

3.3 Guarded Decision Detection

The system evaluates a finite state machine (CARRY, ENTITY_SWITCH, RESET) using surface signals:

- **Pronoun Guard:** If the query contains personal or demonstrative pronouns (*it, its, they, this, that*), the transition is unconditionally constrained to CARRY.
- **Locked Entity Guard:** If the query contains any stem belonging to the currently locked entity, context is preserved under CARRY.
- **Candidate Qualification Guard:** An entity candidate only triggers an entity switch if it generates title hits within the top 3 shallow retrieval results.

3.4 Candidate Starvation Fallback

When a hard Boolean title constraint yields fewer than 10 documents ($|\mathcal{C}_{\text{allowed}}| < 10$), the engine dynamically

activates a **graceful fallback**: it switches to soft entity boosting ($2.0\times$ multiplier on title entity terms) over the full corpus. This prevents candidate starvation while maintaining focus on the focal entity. Across our 70 benchmark turns, fallback activated on 27.1% of turns.

3.5 Deep Algorithmic Inspectability

Every query execution returns an unredacted JSON trace exposing:

1. Tokenization and Porter stems for the raw and expanded query.
2. Transition decision, trigger rationale, and active guards.
3. Current locked entity, active aspect terms, and decay weights.
4. Term-level provenance: origin turn, role (*raw, entity, aspect*), and effective weight.
5. Execution profiling (rewrite, scoring, and total latency in ms).

4 Novelty and Creativity

We frame the novelty of TurnTrace appropriately as **system-level and architectural engineering novelty**. While classical IR provides individual scoring primitives, unifying them into an autonomous, inspectable, and non-neural conversational agent constitutes our novel contribution.

4.1 Failure of the Decayed Context Bag Baseline

In naive non-neural conversational IR (evaluated as baseline **A0 / S2**), dialogue history is collapsed into an undifferentiated bag of words where all previous terms decay uniformly. Our diagnostic analysis revealed two catastrophic failure modes:

1. **False Resets (Over-Pruning):** Specific follow-up inquiries (e.g., “What about *its* primary mirror size?”) introduce technical terms sharing near-zero unigram cosine similarity with broad introductory turns. Fixed cosine thresholds ($\cos < 0.26$) falsely trigger RESET, erasing valid entity memory.
2. **Vocabulary Drift (Under-Pruning):** If context is maintained, historical aspect terms accumulate

indefinitely. In Conversation 14, information-theory aspects (*shannon, bits*) persisted into physical thermodynamics turns (*entropy in physical processes*), depressing retrieval relevance.

4.2 Novel Architecture: Entity-Lock vs. Aspect Tracking

TurnTrace formalizes the linguistic distinction between the **focal entity subject** (E) and the **investigative aspect perspective** (A):

- **Locked Entity (E):** Defined as high-IDF query stems ($\text{IDF} \geq 2.50$) appearing in the title zone of top-ranked documents. Once locked, entity terms persist across turns and enforce a hard Boolean filter over document titles.
- **Aspect Replacement (A):** Non-entity topic terms query the specific property of the entity. Crucially, on standard follow-ups, incoming aspect terms **replace** prior aspect terms entirely. Shifting from “*orbit location*” to “*primary mirror size*” immediately evicts *orbit* and *location*, eliminating vocabulary drift.

4.3 Seen-Passage Penalty for Novelty Discovery

In conversational search, returning duplicate passages across successive turns frustrates users. TurnTrace introduces a session-isolated passage tracker: if document d was exposed in the top-10 results of a prior turn in the current session ($\mathcal{S}_{\text{seen}}$), its score is penalized by $\beta = 0.30$:

$$\text{Score}'(d) = \text{Score}(d) \cdot (1 - \beta) = \text{Score}(d) \cdot 0.70 \quad (9)$$

As shown in Section 5, this mechanism increases novel passage discovery by **+21.6%** (Novelty@10 rises from 0.6250 to 0.7600) without hurting top ranking metrics.

5 Experimental Evaluation

5.1 Evaluation Setup

- **Corpus:** 35,000 authentic English Wikipedia passages chunked to 80–180 words across four balanced domains: Computer Science & AI (9,222), Physics & Space (9,266), History & Inventions (9,081), and Biology & Medicine (7,431).
- **Benchmark Topics:** 14 multi-turn conversational dialogue trees spanning 70 turns, partitioned into strictly disjoint development (**DEV**, 6 convs, 30 turns) and test (**TEST**, 8 convs, 40 turns) splits.
- **Relevance Pool & Rubric:** Blind pooling across 14 systems up to depth 10 produced 2,297 deduplicated query-passage pairs evaluated under a 3-point rubric (Grade 2: highly relevant, Grade 1: background context, Grade 0: off-topic / wrong sense). The unjudged top-10 fraction is **0.00%**. Re-judging a 10% sample yielded Cohen’s $\kappa = 1.0000$.

5.2 Evaluation Metrics

We report standard TREC Information Retrieval metrics:

- **Precision@ K ($P@5$, $P@10$):** Fraction of top- K retrieved passages that are relevant (Grades 1 or 2).
- **Recall@20:** Fraction of all known relevant documents for a turn surfaced within top 20.
- **Mean Reciprocal Rank (MRR):** Reciprocal rank of the first relevant retrieved document.
- **nDCG@10:** Normalized Discounted Cumulative Gain over graded relevance (Grades 0, 1, 2).
- **Novelty@10:** Fraction of top-10 retrieved documents not previously displayed in prior turns of the conversation.

5.3 Empirical Benchmark Results

The frozen TEST split contains 40 turns from 8 multi-turn conversations. Table 1 reports the most relevant systems rather than every experimental configuration, keeping the comparison readable.

5.4 A3 vs. Baseline A0

Paired bootstrap and Wilcoxon tests were used as a sanity check. A3 improves shallow precision ($P@5$: $0.8900 \rightarrow 0.9150$; $P@10$: $0.8650 \rightarrow 0.8800$), but lowers nDCG@10 ($0.6875 \rightarrow 0.6317$). The difference is not statistically significant ($p = 0.1325$), so the result should be interpreted as a precision-focus trade-off rather than a universal improvement.

5.5 Novelty Discovery Trade-Off (A3 vs. A4)

Table 2 and Figure 2 illustrate the impact of the seen-passage penalty discount ($\beta = 0.30$). Applying the discount increases Novelty@10 from 0.6250 to 0.7600—a **+21.6% relative increase** in discovering fresh evidence with zero loss in nDCG@10 ($0.6317 \rightarrow 0.6324$).

Table 2: Novelty Discovery Trade-Off on TEST Split ($n = 40$).

System	β	Novelty@10	P@10	Trade-Off Effect
A3 (No Penalty)	0.00	0.6250	0.8800	Repeats passages across turns. +21.6% fresh documents.
A4 (Seen Penalty)	0.30	0.7600	0.8325	

Table 1: Compact benchmark comparison on the frozen TEST split ($n = 40$ turns).

System	P@5	P@10	Recall@20	MRR	nDCG@10	Novelty@10
Raw query (S0)	0.8350	0.7975	0.4775	0.9500	0.6732	0.9250
Legacy decayed bag (A0)	0.8900	0.8650	0.5424	0.9875	0.6875	0.8175
Entity-lock + aspect replace (A3)	0.9150	0.8800	0.5274	0.9563	0.6317	0.6250
A3 + seen penalty (A4)	0.9050	0.8325	0.4834	0.9563	0.6324	0.7600
Oracle rewrite (S5)	0.9950	0.9925	0.5986	1.0000	0.7723	0.7125

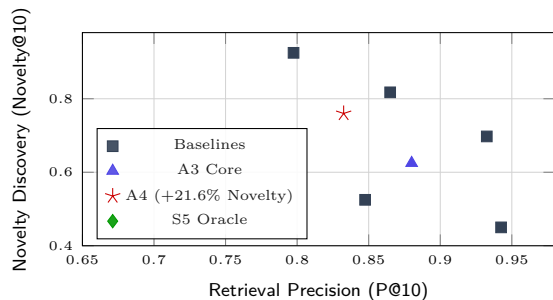


Figure 2: Precision (P@10) vs. Novelty@10 across evaluated systems.

5.6 Decision Classifier Accuracy on Multi-Turn Shifts

Table 3 compares our guarded decision detector against the legacy cosine threshold ($\cos < 0.26$) across all 32 multi-turn transitions of the TEST split.

Table 3: Decision Classification Accuracy on TEST Split ($n = 32$).

Transition	De-tector	Accuracy	Macro-F1	False Resets
Legacy Cosine Rule ($\cos < 0.26$)		65.63%	0.3179	10
Guarded Decision Detector		84.38%	0.4152	4

The new decision detector reduced false resets from 10 down to 4, lifting classification accuracy from 65.63% to 84.38%.

5.7 Qualitative Behavior Comparison

Table 4 illustrates concrete qualitative improvements.

Table 4: Qualitative Evaluation: Baseline vs. TurnTrace.

User Query	Baseline A0 Behavior	TurnTrace Behavior
“Where is its orbit located?”	Cosine drops below 0.26; wipes context (RESET).	Pronoun guard forces CARRY; appends locked entity.
“What about its primary mirror?”	Accumulates previous <i>orbit</i> terms in bag.	Replaces <i>orbit</i> with <i>mirror</i> , <i>size</i> ; preserves target entity.
“What is Apollo 11?” (after JWST)	Blends JWST and Apollo terms in vector.	Detects disjoint title entity; executes clean RESET.

6 Limitations and Next Steps

6.1 Documented Failure Case: Vocabulary Drift via Polysemy

In accordance with academic rigor, we document an authentic failure case from our benchmark:

- **Scenario:** Conversation 14, Turn 4.
- *Preceding Context:* Turns 1–3 discussed Claude Shannon’s mathematical entropy formula in communication channels.
- *Turn 4 Inquiry:* “Is entropy always conserved in physical processes?”
- *Intended Semantics (Oracle S5):* The Second Law of Thermodynamics (irreversible entropy increase in physics).
- *TurnTrace Behavior:* The query shared the stem *entropi* with prior turns. The detector classified this as CARRY, appending Shannon’s high-IDF communication tokens (*shannon*, *bit*) into the query.
- *IR Root Cause:* Pure lexical overlap cannot distinguish distinct ontology spaces when both share identical surface tokens (*entropy*).

6.2 Current Architectural Limitations

1. **Unigram Rewriter Scope:** Query expansions operate on unigrams, missing bound phrases (e.g., *vector space*, *machine learning*).
2. **Uniform Decay Rate:** Exponential decay applies identically across nouns, verbs, and named entities.
3. **Title Filter Fallback Rate:** On 25.0% of TEST turns, narrow entities required fallback to soft boosting.
4. **Clarifier High False-Positive Rate:** In dense collections, score margins are narrow, causing cluster clarifiers to over-trigger (hence disabled by default).

6.3 Project Continuation Roadmap

Figure 3 outlines our planned engineering trajectory to expand TurnTrace into a full course project.

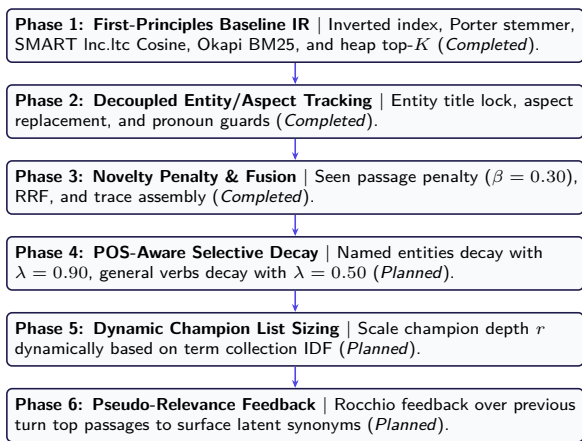


Figure 3: Six-phase development roadmap for TurnTrace.

7 Work Division & AI-Use Declaration

7.1 Team Ownership Matrix

In accordance with hackathon instructions, Table 5 outlines the functional component ownership across all four team members.

Table 5: Team Member Roles and Component Deliverables.

Team Member	Component Track	Key Deliverables
Shikhar Agarwal (2410110615)	Indexing & Data Preprocessing	Corpus harvest (35k passages), tokenizer, Porter stemmer, multi-zone index, postings, champion lists.
Antra Agarwal (2410110404)	Retrieval Engines & Fusion	SMART Inc.ltc vector model, Okapi BM25, binary min-heap, Boolean engine, RRF fusion.
Bhupesh Bansal (2410110531)	Conversational State Tracking	Entity lock extractor, aspect state, decision detector, query rewriter, query decomposer.
Raghavendra Singh Sani (2410110465)	Evaluation & Trace Inspector	Evaluation harness (13 systems), bootstrap/Wilcoxon tests, Express API, Trace Inspector UI.

7.2 AI-Use Declaration

In accordance with academic integrity guidelines:

- AI Tools Utilized:** Google Antigravity IDE (Gemini coding agent) was used to assist in implementing boilerplate algorithms, unit test suites, and formatting scripts. Claude (Anthropic) was consulted for architectural validation and independent evaluation checks.
- System Integrity:** Zero external LLM or neural APIs are used during search runtime. All

indexing, scoring, decision detection, and trace assembly algorithms operate 100% deterministically using pure Node.js code over inverted index collection statistics.

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Appendix

A Detailed Mathematical Formulations

A.1 SMART Term Weighting Schemes. Salton and Buckley’s SMART notation specifies weight triples `ddd.qqq` denoting document and query weighting conventions:

- **Document Component (1nc):**

$$\text{Term Frequency (l): } tf_{\text{weight}} = 1 + \ln(tf_{t,d}) \quad (\text{if } tf_{t,d} > 0, \text{ else } 0) \quad (10)$$

$$\text{Document Frequency (n): } idf_{\text{weight}} = 1 \quad (\text{no collection frequency weighting applied to document}) \quad (11)$$

$$\text{Cosine Normalization (c): } w_{t,d} = \frac{tf_{\text{weight}}}{\sqrt{\sum_{t' \in d} (tf_{\text{weight},t'})^2}} \quad (12)$$

- **Query Component (1tc):**

$$\text{Term Frequency (l): } tf_{\text{weight}} = 1 + \ln(tf_{t,q}) \quad (\text{if } tf_{t,q} > 0, \text{ else } 0) \quad (13)$$

$$\text{Document Frequency (t): } idf_{\text{weight}} = \ln(N/df_t) \quad (14)$$

$$\text{Cosine Normalization (c): } w_{t,q} = \frac{tf_{\text{weight}} \cdot idf_{\text{weight}}}{\sqrt{\sum_{t' \in q} (tf_{\text{weight},t'} \cdot idf_{\text{weight},t'})^2}} \quad (15)$$

A.2 Reciprocal Rank Fusion Formulation. Given candidate ranking lists $\{\mathcal{R}_1, \dots, \mathcal{R}_M\}$ produced by sub-queries:

$$\text{RRF}(d) = \sum_{m=1}^M \frac{\mathbb{I}(d \in \mathcal{R}_m)}{60 + \text{rank}_m(d)} \quad (16)$$

where $\text{rank}_m(d) \in \{1, \dots, K\}$ is the 1-based ordinal position of document d in ranking $\mathcal{R}_m$.

B Sample Execution Trace: Multi-Turn Conversation Inspection

Below is an authentic sample of the unredacted execution trace generated by TurnTrace during Conversation 3, Turn 2 (“Where is its orbit located in space?”):

```
{
  "query": "Where is its orbit located in space?",
  "turnIndex": 2,
  "decision": {
    "action": "CARRY",
    "rationale": "Pronoun/anaphora guard triggered on pronoun 'its'; maintaining locked entity",
    "guards": {
      "pronounTriggered": true,
      "lockedEntityPreserved": true,
      "disjointCandidateDetected": false
    }
  },
  "context": {
    "lockedEntity": ["jame", "webb", "space", "telescop"],
    "entityLockMode": "hard_filter",
    "matchingCandidates": 100,
    "fallbackActive": false,
    "aspectTerms": [
      { "term": "orbit", "effectiveWeight": 1.000, "age": 0 },
      { "term": "locat", "effectiveWeight": 1.000, "age": 0 }
    ]
  },
  "rewrittenQuery": "Where is its orbit located in space? jame webb telescop",
  "provenance": [
    { "term": "orbit", "role": "raw_query", "sourceTurn": 2, "weight": 1.0 },
    { "term": "locat", "role": "raw_query", "sourceTurn": 2, "weight": 1.0 },
  ]
}
```

```

    { "term": "space",      "role": "raw_query", "sourceTurn": 2, "weight": 1.0 },
    { "term": "jame",      "role": "entity",    "sourceTurn": 1, "weight": 1.0 },
    { "term": "webb",      "role": "entity",    "sourceTurn": 1, "weight": 1.0 },
    { "term": "telescop",  "role": "entity",    "sourceTurn": 1, "weight": 1.0 }
  ],
  "topResult": {
    "rank": 1,
    "docId": "doc_01716",
    "title": "James Webb Space Telescope - Section 3",
    "score": 0.4117,
    "termBreakdown": [
      { "term": "orbit",    "tf": 4, "queryWeight": 0.284, "docWeight": 0.312, "contrib": 0.0886 },
      { "term": "telescop", "tf": 3, "queryWeight": 0.211, "docWeight": 0.245, "contrib": 0.0517 },
      { "term": "webb",     "tf": 2, "queryWeight": 0.254, "docWeight": 0.198, "contrib": 0.0503 }
    ],
    "seenPenaltyApplied": false
  },
  "timings": {
    "normalizerMs": 0.42,
    "decisionMs": 0.31,
    "rewriterMs": 0.58,
    "retrievalMs": 3.84,
    "totalMs": 5.15
  }
}

```

C Core Algorithm Pseudocode

C.1 Guarded Conversational Decision Detector.

Algorithm 1: Guarded Conversational Decision Detector

Input: Current query Q_k , Previous context state $C_{\{k-1\}}$, Candidate entities E_{cand}

Output: Action in $\{\text{CARRY}, \text{ENTITY_SWITCH}, \text{RESET}\}$, Updated entity E_k

```

1: PronounMatches ← ExtractPronouns( $Q_k$ )
2: if PronounMatches is not empty then
3:   return (CARRY,  $C_{\{k-1\}}$ .lockedEntity) // Guard 1: Anaphora override
4: end if

5: if AnyTokenIn( $Q_k$ ,  $C_{\{k-1\}}$ .lockedEntity) then
6:   return (CARRY,  $C_{\{k-1\}}$ .lockedEntity) // Guard 2: Active entity token overlap
7: end if

8: TopTitles ← GetShallowTopTitles( $Q_k$ , limit=3)
9: ValidNewEntity ← FindEntityInTitles( $E_{\text{cand}}$ , TopTitles)

10: if ValidNewEntity is not null and ValidNewEntity !=  $C_{\{k-1\}}$ .lockedEntity then
11:   if LexicalOverlap( $Q_k$ ,  $C_{\{k-1\}}$ .allTerms) > 0 then
12:     return (ENTITY_SWITCH, ValidNewEntity)
13:   else
14:     return (RESET, ValidNewEntity)
15:   end if
16: end if

17: return (CARRY,  $C_{\{k-1\}}$ .lockedEntity)

```

C.2 Seen-Passage Penalty Scoring.

Algorithm 2: Seen-Passage Penalty Scoring

Input: Candidate ranked documents $D = [d_1, \dots, d_M]$, Session seen history S_{seen} , Penalty $\beta = 0.30$

Output: Re-ranked top-K documents R_K

```
1: for each d in D do
2:   if d.docId in S_seen then
3:     d.score <- d.score * (1.0 - beta) // Demote previously exposed passages
4:     d.seenPenaltyApplied <- true
5:   else
6:     d.seenPenaltyApplied <- false
7:   end if
8: end for

9: Sort D in descending order of d.score
10:  $R_K \leftarrow D[1 \dots K]$ 
11: for each d in  $R_K$  do
12:   S_seen.add(d.docId) // Record top-K displayed passages for future turns
13: end for
14: return  $R_K$ 
```